CERN Bulletin

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CERN and the US sign joint statement of intent

Joint Statement of Intent between the US and CERN concerns future planning for large research infrastructures, advanced scientific computing and open science

CERN Director-General, Fabiola Gianotti (right), and Principal Deputy US Chief Technology Officer, Deirdre Mulligan, of the White House Office of Science and Technology (left) at the signing ceremony. (Image: US Department of State, Bureau of Oceans and International Environmental and Scientific Affairs)

CERN and the US government have released a joint statement concerning future planning for large research infrastructures, advanced scientific computing and open science. The Joint Statement of Intent was signed in Washington DC in April by CERN Director-General, Fabiola Gianotti, and Principal Deputy US Chief Technology Officer, Deirdre Mulligan, of the White House Office of Science and Technology (pictured).

Acknowledging their longstanding partnership in nuclear and particle physics, CERN and the US intend to enhance collaboration in planning activities for large-scale, resource-intensive facilities with the goal of providing a sustainable and responsible pathway for the peaceful use of future accelerator technologies.

Concerning the proposed Future Circular Collider, FCC-ee, which would collide electrons and positrons to produce copious quantities of Higgs bosons, the text states: “Should the CERN Member States determine the FCC-ee is likely to be CERN’s next world-leading research facility following the high-luminosity Large Hadron Collider, the United States intends to collaborate on its construction and physics exploitation, subject to appropriate domestic approvals.”

Contents

A technical and financial feasibility study for the proposed FCC is due to be completed in March 2025.

CERN and the US also intend to discuss potential collaboration on pilot projects to incorporate new analytics techniques and tools such as AI into particle physics research at scale and affirm their collective mission "to take swift strategic action that leads to accelerating widespread adoption of equitable open research, science, and scholarship throughout the world".

In December 2023, the high-energy physics advisory panel to the US Department of Energy and the National Science Foundation released a 10-year strategic plan for US particle physics. Meanwhile, the next update of the European Strategy for Particle Physics, which is formed through a broad consultation of the particle physics community in Europe and beyond, is about to get under way. The CERN Council has set the deadline for submitting written input for the next Strategy update at 31 March 2025, with a view to concluding the process in June 2026. The final report of the FCC Feasibility Study will be a key component of that input.

Matthew Chalmers

Accelerator Report: Exploring potential performance increases

Over the years, the teams responsible for the LHC proton injector chain (Linac 4, PS Booster, PS and SPS) have developed various production schemes for the LHC beam, pushed the performance of the beam and explored its potential to enhance the collisions in the LHC. In 2023 and this year, until the end of last week, the so-called "standard LHC beam" has been used in batches of 3 x 36 bunches, provided by the SPS. On 24 May, the LHC was switched to the "BCMS (Beam Compression, Merging and Splitting) beam" mode to explore its potential to produce more collisions and to compare its performance to that of the standard beam.

In the LHC injector chain, the standard beam is produced by injecting three bunches from the PS Booster into the PS. After an initial acceleration, the PS splits each bunch longitudinally (see box) into three, resulting in nine bunches. These nine bunches are then accelerated to the maximum energy of the PS, where each bunch is split into two, and then again into two, resulting in 36 bunches, each spaced by 25 ns (see Figure 1).

The SPS receives three of these 36-bunch shots from the PS and accelerates them to an energy of 450 GeV before injecting them in the clockwise or counter-clockwise direction into the LHC. This means that one PS Booster bunch results in 12 bunches in the LHC. The number of protons per bunch (named intensity) required by the LHC is 16×10^{10}.

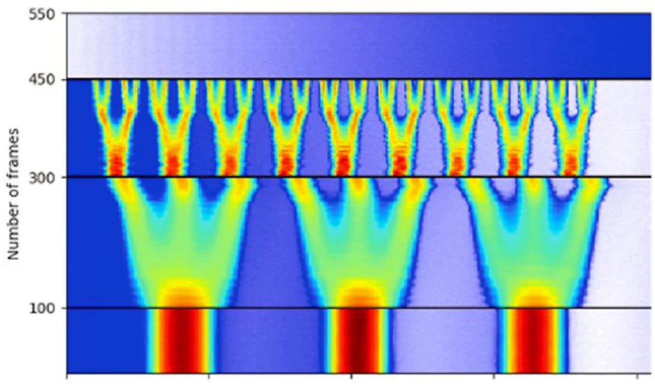


Figure 1: The standard production scheme. The three bands at the bottom of the diagram represent the three PS Booster bunches injected into the PS. The middle band shows the splitting into three, while the top band shows the double split into two, which results in 36 bunches. (Image: CERN)

Taking the 12-fold splitting into account, this means that the number of protons per bunch which the PS Booster has to inject into the PS is 12 times higher than the LHC bunch intensity, i.e. 192×10^{10} protons per bunch.

The BCMS beam is produced by injecting six bunches into the PS: three from a first cycle and three, 1.2 seconds later, from a second cycle. After an initial acceleration, these six bunches are compressed and merged, in pairs of two, into a single bunch, resulting in three bunches, which are then each split into three bunches. The remainder of this production scheme is identical to the standard production scheme, which also results in 36 bunches spaced by 25 ns. With this scheme, six bunches are manipulated to obtain 36 bunches, which gives a splitting factor of six. Therefore, to obtain a bunch intensity of 16 x 10^{10} protons for the LHC, the PS Booster needs to provide only 96 x 10^{10} protons per bunch (see Figure 2).

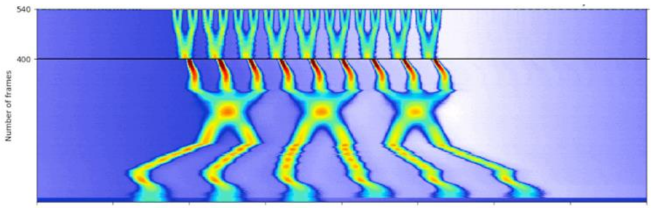


Figure 2: The BCMS production scheme. The six bunches injected from the PS Booster can be seen at the bottom of the diagram. These bunches are compressed in pairs of two and then merged into three bunches, after which each bunch is split into three. In the top part of the image, the same double split into two is applied, as in the standard production scheme, resulting in 36 bunches. (Image: CERN)

The LHC has now used the BCMS beam for about a week and the first signs of improved performance compared with the standard beam have already been observed.

How is it that the BCMS beam results in more collisions in the LHC if it contains the same number of protons as a standard beam?

The BCMS beam has a greater brightness, which means that it contains the same number of protons but in a smaller beam size. This smaller beam size is the result of the lower intensity per bunch in the PS Booster.

The challenge is to preserve this increased brightness when the beam is accelerated in all the machines of the LHC injector chain and in the LHC itself. During acceleration in the LHC, the beam size seems to increase slightly more with the BCMS scheme than with the standard beam scheme. Studies of the beam behaviour and adjustments of the machine parameters may limit this growth in the future, further increasing the number of collisions.

Final adjustments will be made in the coming weeks. A fact-based comparison will allow us to decide whether to continue using the BCMS production scheme or to revert to the standard production scheme. Stay tuned!

Bunch splitting, an explanation:

In the world of particle accelerators, we focus on two main spatial dimensions: transverse and longitudinal.

- The transverse plane refers to the horizontal (left-right) and vertical (up-down) movements of the particles. When we talk about transverse beam size, we measure how wide and tall the beam is in these directions.
- The longitudinal plane is the plane along the path of the accelerator, used to measure the length of the bunches and the spacing between them.

Bunch splitting refers to splitting a single bunch of particles into two or three shorter bunches along the longitudinal plane. The transverse size of the individual bunches remains unchanged.

Rende Steerenberg

CERN70: The world's first hadron collider

Part 10 of the CERN70 feature series. Kjell Johnsen was the Intersecting Storage Rings (ISR) project leader when the accelerator was built

On 27 January 1971, the world's first collisions between two beams of protons occurred in CERN's Intersecting Storage Rings (ISR).

By the late 1950s, physicists knew that a huge gain in collision energy would become possible by colliding proton beams head on, rather than using a single beam and a stationary target. However, the price would be relatively low reaction rates and less variety than those produced at a "fixed target" machine.

The particle physicists at CERN mainly worked on fixed-target accelerators, such as the Proton Synchrotron (PS), and they favoured an upgrade of the PS and the construction of the Super Proton Synchrotron (SPS)

over building a proton–proton collider. Nevertheless, the accelerator physicists prevailed and the ISR was approved in the December 1965 Council session. Construction began the following year.
The ISR proved to be an excellent instrument for particle physics producing many important results, which have become part of the general background knowledge in particle physics. However, research in accelerator studies was at least as important and led to many technological inventions and developments; for example, the ISR saw the world's first proton–antiproton collisions, as well as heavy-ion collisions.
The ISR paved the way for future hadron colliders, from the conversion of the SPS into a proton–antiproton collider from 1981 to 1991, which led to the discovery of W and Z bosons in 1983, to the Large Hadron Collider (LHC) and its fabulous discovery of the Higgs boson in 2012.

Recollections

Kjell Johnsen was head of the ISR department and ISR project leader when the accelerator was built.

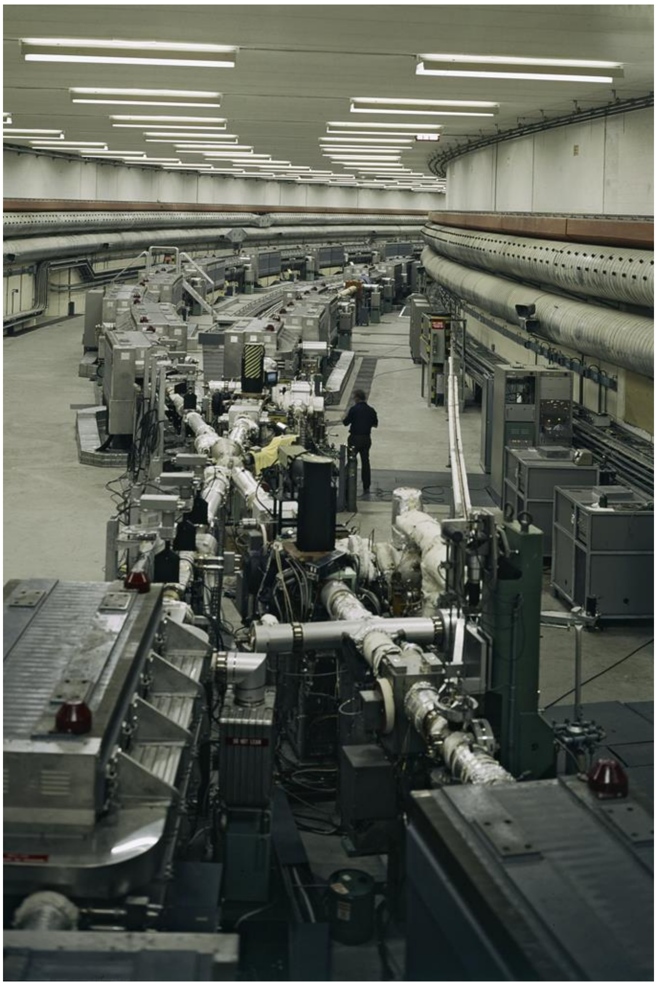
A 1973 photo of an intersection region of the Intersecting Storage Rings (ISR), where data on proton–proton interactions in the highest energies available were being collected. (Image: CERN)

"There are many stories I could tell, but the one I have chosen sums up well the interaction between man and machine that characterised the ISR. Like all pioneering particle accelerators, the ISR was its own prototype. As a result, we were learning until the last, and we came to see the ISR almost as a sentient thing, a sometimes-difficult colleague even.
Let me return to June 1984, when the word came from CERN's research board that the ISR was to close. We received the instruction that the beam that was in the machine would be the last, but that the ISR would keep running for as long as we could keep that beam. The machine reflected on this for a moment, then produced a good solid hiccup and spat out the few antiprotons it had circulating. We decided that we couldn't let the ISR have the final word, and in due course we were granted one more fill. We kept it circulating, and doing good physics, for about 100 hours. The ISR was behaving impeccably until 6 am on 25 June, when we switched it off, one day before the official closing ceremony.

On 27 January 1971, beams of protons collided in the Intersecting Storage Rings (ISR) for the first time. Kjell Johnsen (centre) shows Willi Jentschke and Hildred Blewett some emerging results. (Image: CERN)

On that occasion, Viki Weisskopf, CERN's Director-General at the time the ISR project was approved, summed up the ISR's achievement with the words: "The really important thing about the ISR is its success as an instrument, because that fact did change the landscape of high-energy physics. [...]

It showed the possibility of doing high-energy physics at much higher energies in the centre-of-mass system. [...] After this was done, colliders became the fashion of the day."



A 1974 photograph of the ISR tunnel at one of the intersection points (I5) showing the layout of the magnets and the crossing of the two beam pipes. (Image: CERN)

As an accelerator physicist, I find it appropriate to add that this success came thanks to a number of firsts in accelerator physics. The ISR's first – and main – aim was to collide proton beams and it did this brilliantly over its 13 years lifespan, and with ever increasing luminosity. Of the other firsts, it was at the ISR that stochastic cooling was proven, with antiproton beams being kept for hundreds of hours. It was also at the ISR that protons and antiprotons first collided, and the ISR was the first heavy-ion collider, colliding alpha particles. Ultra-high vacuum techniques, necessary for colliders, came of age at the ISR and it was at the ISR that superconducting magnets were first used with circulating beams.

Many feel that the ISR was switched off in its prime, and this may be true. However, in a world of limited resources, priorities have to be set and the particle physics community decided that the ISR had done its job. To close, I would again like to echo the sentiments of Viki Weisskopf, who pointed out at the closing ceremony that children frequently outshine their parents. As the first hadron collider, the ISR can count among its offspring illustrious machines such as the SPS proton–antiproton collider at CERN, the Tevatron at Fermilab, RHIC at Brookhaven, and perhaps even HERA at DESY. [...] The ISR therefore has pride of place in the history of particle physics as founder of an illustrious dynasty."

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. Kjell Johnsen passed away in 2007 at the age of 86, read a tribute to him in the CERN Courier as well as a feature on the ISR, also in the CERN Courier.

HiLumi News: The HL-LHC's cold powering system successfully passed the tests

The HL-LHC cold powering system undergoing tests in SM18. (Image: CERN)

If you're an avid follower of High-Luminosity LHC (HL-LHC) news, you will no doubt already have heard about "the python", the new superconducting link developed at CERN. It is a component of the new cold powering system that will power the HL-LHC inner triplet magnets, which will focus proton beams more tightly around the ATLAS and CMS collision points.

This new system is packed with novel superconducting technologies: MgB_2 superconducting cables, twisted together to form a compact bundle of about 9 centimetres in diameter, are inserted into a 22-centimetre-diameter flexible cryostat, with vacuum insulation and flowing helium gas. The MgB_2 cables operate in the helium gas at temperatures from about 4.5 K (-268.7 °C) to 20 K (-253.2 °C). The REBCO high-temperature superconducting cables then transfer the current from 20 K to 50 K (-223.2 °C) and, finally, current leads provide the transition from 50 K to room temperature. This system can carry a direct electrical current (DC) of around 120 kA over the required distance of 85 metres.

While the superconducting cables of the LHC magnets have to be maintained in superfluid helium (at 1.9 K (-272.2 °C)) or in liquid helium (at 4.5 K), the new superconducting part of the system is capable of operating at a temperature of up to 60 K (-213.2 °C) at its highest, qualifying it as "high temperature" in superconductivity terms. "One of the beauties of this new system is that it operates in helium gas. The cryogenic cooling of the superconducting link is at zero cost, because it transfers the helium gas that in any case is needed to cool the current leads. This is one of the benefits of using high-temperature superconductors," explains Amalia Ballarino, leader of the HL-LHC Cold Powering Work Package.

The superconducting link and its flexible cryostat can be spooled onto a large drum and transported like conventional power transmission cables. This new type of superconducting system has enormous potential for future accelerators and in areas beyond accelerator technology where large transfer of current is needed, or for the development of clean aviation.

The new superconducting links will connect the power converters, located in radiation-free underground technical galleries above the LHC tunnel, to the HL-LHC magnets. The distance between the two link ends spans about 85 m for the inner triplets and includes a vertical path via an 8-m shaft (simulated here by the ramp visible in the photo). (Image: CERN)

The first HL-LHC cold powering system has just passed its first tests in the SM18 test facility. While the python was fully qualified in the previous R&D phases, this is the first time that a full power transmission system, transferring current from room temperature to the liquid helium environment via MgB_2 and REBCO superconducting technology, has been constructed and successfully validated in final operating conditions. The complexity of the system is enhanced by the multiplicity of the circuits it contains. “The 19 superconducting cables and current leads, rated at currents ranging from 2 kA to 18 kA, transported a total DC current of 94 kA, the maximum current that could be delivered by the test station,” adds Ballarino. “Electromagnetic compatibility among circuits was validated, and high-voltage insulation tests were successfully accomplished. This great success is the result of ten years of R&D.”

The next steps will take place in early summer, when the cold powering system will be transported to the HL-LHC IT String where the collective behaviour of the inner triplet magnet system will be tested prior to installation underground in the LHC during the next long technical stop (LS3), scheduled to begin in 2026.

To find out more, read this article published in the CERN Courier in April 2023.

Anaïs Schaeffer

NA64 uses the high-energy SPS muon beam to search for dark matter

The NA64 collaboration has used a muon beam for the first time to search for dark matter and could set new limits for a potential dark Z' boson

The setup of the NA64 experiment in the SPS muon beamline (called the M2 beamline). (Image: CERN)

The NA64 experiment started operations at CERN’s SPS North Area in 2016. Its aim is to search for unknown particles from a hypothetical “dark sector”. For these searches, NA64 directs an electron beam onto a fixed target. Researchers then look for unknown dark sector particles produced by collisions between the beam’s electrons and the target’s atomic nuclei.

Recently, the NA64 team started using a muon beam from the SPS to search for new particles that interact predominantly with muons – heavier versions of the electron – and could explain simultaneously the long-standing puzzle of the muon’s anomalous magnetic moment and the dark-matter (DM) problem. Their first results were accepted in the journal Physical Review Letters on 8 April.

In this paper, the NA64 collaboration sets new limits on the available parameter space – the window where the researchers could find a hypothetical dark boson Z’ coupling only to muons and tauons for given values of its mass and coupling strength. In the so-called vanilla model, the Z’ can only decay back into neutrinos and could provide an explanation of the muon’s anomalous magnetic moment puzzle. However, in extended models, it can also decay into DM candidates. This would solve the DM problem by predicting the observed relic density of DM particles created in the early universe. With these results, the NA64 collaboration demonstrates the great potential of muon beams in dark matter searches and in future new physics scenarios preferably coupled to muons.

“Muons scattering off the nuclei in the target could produce a hypothetical dark boson Z’, followed by its invisible decay into either a pair of neutrinos or a pair of dark-matter candidates, depending on the underlying model,” explains the deputy Technical Coordinator Laura Molina

Bueno. “The signature of this production would be missing energy and momentum in our detectors.”

To search for this, a 160 GeV tertiary muon beam derived from the primary SPS proton beam is fired onto an electromagnetic calorimeter acting as an active target. The experimentalists then search for events in which one final-state muon has a momentum lower than 80 GeV with no detectable activity in the downstream calorimeters.

As no event matching these conditions was observed in the expected signal region, the researchers were able to exclude this region and conclude that, for the first model, the only possible mass window for a dark boson Z’ to explain the g-2 muon anomaly is from 6 MeV up to 40 MeV. Their results also indicate that light thermal dark matter coupled to the standard model via a (Lmu-Ltau) Z’ cannot be heavier than 40 MeV.

NA64 is among the first experiments searching for dark sectors weakly coupled to muons. The experimentalists are confident that they will cover the available parameter space in the future by using higher beam intensities. “Using a muon beam opens a new window to explore other well-motivated new physics scenarios, such as benchmark dark-photon models, scalar portals, millicharged particles or lepton-flavour violating processes,” concludes NA64 co-Spokesperson Paolo Crivelli.

Kristiane Bernhard-Novotny

The CERN Alumni Network turns seven

Following the successful Third Collisions event in February, the Network will host a LinkedIn live event on 13 June to celebrate its seventh anniversary

2024 is proving to be an exhilarating year for the CERN Alumni Network, which will turn seven on 8 June. Join the seventh-anniversary LinkedIn live on Thursday, 13 June to discover the impactful work being carried out by its members and celebrate our shared achievements together.

The CERN Alumni Network, which boasts nearly 10 000 members, is an integral part of the CERN community as it enables alumni to keep in touch with the Organization and each other after leaving the Laboratory. Throughout the year, the Network organises events to connect with alumni and companies and it also offers career guidance and mentoring.

Between 9 and 11 February 2024, CERN witnessed the momentous gathering of just under 600 people for the Network’s triennial reunion, Third Collisions. This vibrant event served as a testament to the enduring camaraderie within the CERN community, providing a platform for alumni to reconnect and exchange ideas. From captivating keynote addresses to interactive panel discussions, participants explored a wide array of topics, showcasing the breadth of knowledge within the alumni network and its collective commitment to addressing global challenges. Recordings of many of the keynote talks and parallel sessions can be accessed on the Indico event page. One highlight of Third Collisions was the inclusion of the first careers fair, which provided a platform for companies to connect directly with alumni. Networking sessions and CERN Club activities further reinforced the sense of a CERN community at the event. Participants also had the privilege of exploring the newly inaugurated Science Gateway, which served as a fitting backdrop for discussions on cutting-edge research and innovation.

Third Collisions was more than just a reunion: it was a convergence of minds, ideas and experiences. Energised by new insights and connections, participants departed with a renewed sense of purpose and pride in belonging to the extraordinary CERN alumni community. Thanks to Third Collisions, several alumni have come forward to propose new regional groups. Reflecting on the event, one attendee remarked, "I am so proud to be part of such a thriving and inspiring community! All the trajectories of CERN's alumni are super interesting, and it felt like a big family get-together."

If you haven't joined the Network yet, now is the perfect time to do so. By becoming a member, you can expand your professional network, forge connections with individuals who share your CERN experience, participate in exciting events and showcase your ongoing endeavours to a community passionate about the groundbreaking work conducted at CERN. Don't forget to join the LinkedIn live on 13 June to continue these celebrations and connect with this ever-growing community.

CERN Alumni programme

CERN's artists on stage at the Victoria Hall as Fabiola Gianotti receives the 2024 prize from the "Fondation pour Genève"

The prestigious award ceremony, held in Geneva's Victoria Hall, was also an opportunity to celebrate the CERN community

Left: The Canettes Blues Band opening the ceremony; Centre: Fabiola Gianotti, CERN Director-General receiving the prize from Marc Pictet, president of the Fondation pour Genève. Right: The CERN Choir closing the ceremony. (Image: CERN)

On 13 May 2024, members of CERN's vibrant community attended, and some performed at, the prestigious *Fondation pour Genève* prize ceremony at Victoria Hall. Since 1978, the annual prize has honoured Geneva citizens and institutions that contribute to the international influence of the city in scientific, political, economic, cultural and humanitarian fields. CERN received it in 1999. For the 2024 prize, CERN Director-General Fabiola Gianotti was the recipient, honouring her exceptional commitment to the international influence of Geneva.

Musical contributions from the CERN community were at the heart of this celebration, which began with the Canettes Blues Band performing ATLAS Boogie and ended with an excerpt of Niccolò Jommelli's Requiem performed by the CERN Choir. Interspersed throughout the evening were various testimonials, including from CERN community members. Presentations showing CERN's 70-year history and the newly inaugurated Science Gateway, CERN's state-of-the-art centre for education and outreach, celebrated the scientific and cultural impact of CERN in Geneva.

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Watch the full award ceremony on the *Fondation pour Genève* website.

Computer Security: The better generation

To all those fine folks out there who are interested in computer security, who take care of the secrecy of their passwords and other credentials, who protect their laptops and smartphone adequately with up-to-date operating systems and antivirus software, and who apply due diligence when

developing and running their IT services and/or control systems, I would issue just two words:
THANK YOU!
Thank you for reading our articles. Thank you for showing an interest in privacy and security. Thank you for wanting to learn more about this. Thank you, because you are the generation who can get it right. Or better, as my generation of 1971 didn't screw everything up.

What's gone before
Look, for example, at an ancient telephone – the one with a rotary dial. Back then, fear of being spied on was minimal, and only an issue if you annoyed your country. Today, we all carry small spying devices around that collect all our personal information and pass it on. Maybe not immediately to governments, but to big multinationals that make money from our personal data. The secrecy of the post has become WhatsApp, Threema, Signal and Telegram – each with their own privacy-preserving means (or not). With the cloud came the Wild West. Analogue cameras became Instagram and TikTok. Apple revolutionised our record and tape collection. CDs? Bah. MP3s? Not anymore. Linear television became Netflix, Amazon Prime and Disney+. Amazon and Google know much more about our shopping habits than the old neighbourhood shopkeeper ever did. And workout information now goes to Strava, Fitbit or the like. Mapping out the world. Our nicely cloaked private world has become frighteningly transparent and public. Orwell's 1984 surveillance state at its best. At least there is a silver lining in the form of the European Union's General Data Protection Regulation, which the big companies try to aggressively bend and small startups try to creatively circumvent.
Like with privacy, digitalisation over the past decades has tied our lives into symbiosis with technology. Physical security has become cybersecurity. Today, all the amenities of life are technology-supported. Depending where you are, this is the case to varying degrees. Consider electricity. In most of our countries, electricity is the One Ring that rules it all. No electricity, no cold food or (worse) medication. No electricity, no communication. No electricity, no fresh water, as water pumps need electricity. Similarly for fuelling stations. No electricity, no public transport. Going shopping? Erm, how did you pay last time? Of course, you might have some batteries left over, or a diesel generator. But in the long run? We live in symbiosis with a technology backbone. With electricity. With the control systems deployed for running this backbone. In the past, this backbone was threatened only by physical means – by conflicts. By nation states in an increasingly peaceful world. While we thought that those times were gone, our backbone is now much more susceptible to threats. No need for nation states anymore, when a small group of (state-sponsored) criminals can create havoc. Like the attacks on Saudi Aramco. Like Stuxnet against Iranian nuclear centrifuges. Like Russian hackers allegedly attacking Ukrainian infrastructure prior to the invasion of Crimea. Like the ransomware attacks against Maersk. Like the Conti ransomware group against anyone else on this planet. The COVID-19 pandemic and Russia's war against Ukraine have shown how fragile our technological backbone has become, how inherently insecure it is and how easily it can be brought to a halt. Threats to this backbone won't disappear.
And the future, the sunny world of clouds, requires even more backbone. More interconnectivity, more technology, more complexity. Ergo more vulnerabilities. And ergo more severe consequences. Self-driving cars talk to each other and to the traffic lights. Cities become smart. Cashless stores RFID your shopping basket and charge your credit card automatically. Your fridge orders missing items automagically, delivered by drone within 10 minutes. In this brave new Wild West, the genie is out of Pandora's box. Our technological backbone needs reinforcement. The stupid internet of unsecure things needs improvement. The zillions of layers, virtual machines, containers, software interdependencies, agility, DevOps and just-in-time need experts to put the genie back in the bottle. To adapt technology such that it serves but does not burden. To bring security into every single layer by default. Making security an equal among other IT equals: functionality, usability, maintainability, availability and – security. While threats and threat actors will never give up (and will actually become more and more sophisticated), we need to counter the increasing

number of vulnerabilities and keep the consequences of successful attacks at bay.

Now, enter you!

We will never have 100% secure systems – and those who promise this to you are either liars or salespeople or both. “Security will always be exactly as bad as it can possibly be while allowing everything to still function” (Nat Howard). Because we’re lazy and ignorant, because security is usually just a cost factor with limited benefits: security, convenience, cost – pick two. This makes security only as good as the weakest link in the chain of technology. This makes security a people problem. But this also makes security a problem that can be solved by people. You are the crucial generation. The first twists and turns towards a more privacy-preserving and secure future have started. Facebook and Google have been restrained from collecting data. WhatsApp becomes Threema or Signal. Security must again move into focus, joining the other —ities and reinforcing the CIA triangle: confidentiality (hush! for your personal life), integrity (your bank statement) and availability (giving you electricity when you need it). Actually, in industry this is instead the AIC triangle (availability: your supermarket; integrity: the soundness of the bridges you cross to get there; and confidentiality: Coca Cola’s secret recipe).

Since my generation failed to consistently, coherently, efficiently and effectively push those triangles through as it should have, the baton is now handed to you. Together, let’s break up the old mantra of “freedom, security, convenience – choose two” (Dan Geer) and see how we can still get all three deployed on an acceptable level. Open your mind to think secure and privacy-preserving. If you haven’t done so yet, learn how to prevent and protect, how to plan, design, develop and build secure and privacy-preserving applications, software and systems. How to operate systems in a secure and privacy-preserving fashion – finding weaknesses and vulnerabilities, detecting abuse and ensuring that sufficient log information is at hand, and using the magic means available to understand what happened if the evil bad has compromised your system: forensics, incident coordination and response.

In addition to the new round of WhiteHat and Zebra training sessions, which are coming up very soon, we’re happy to announce that dedicated online training courses on all security matters are now available to all of you at any time, with our thanks to the HR training team. The SecureFlag training platform provides hands-on courses, exercises and virtual environments for you to improve your skills in secure software development in your favourite programming language (demo video). Learn how to securely configure your systems, virtual machines and containers and how to securely operate your web and computing services. These new, dedicated courses are provided for your benefit and for the benefit of a secure organisation – to clean up the security and privacy mess. THANK YOU!

The Computer Security team

Do you want to learn more about computer security incidents and issues at CERN? Follow our Monthly Report. For further information, questions or help, check our website or contact us at Computer.Security@cern.ch.

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: https://cern.ch/official-news.

Official News regarding the CERN Safety Rules

The CERN Safety Rules listed below have been published on the CERN website dedicated to the Safety Rules:
Safety Guideline SG-NIR-1-0-1 "Exposure to Static Magnetic Fields"
Safety Guidelines provide detailed instructions for the implementation of CERN Safety Rules. Compliance with a Safety Guideline provides a presumption of conformity with the associated CERN Safety Rule. This Safety Guideline provides guidance for the implementation of General Safety Instruction GSI-NIR-1 "Protection of Persons against Exposure to Static Magnetic Fields".

The CERN Safety Rules apply to all persons under the Director General's authority. They are available under the following link: http://www.cern.ch/safety-rules

HSE unit

Education fees – Period of admissibility

Members of the personnel are reminded that, pursuant to Article R V 1.37 of the Staff Regulations, they have until 31 August 2024 to submit claims for the reimbursement of education fees relating to the 2022/2023 school year.
These claims should be made using an EDH form: https://edh.cern.ch/Document/Claims/EducationFees

Detailed information concerning education fees is available in the Admin e-guide: https://admin-eguide.web.cern.ch/en/procedure/payment-education-fees-summary
The Human Resources department also remains at your disposal to answer any questions: schoolfees.service@cern.ch.

HR department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

27 May–27 June | At CERN | Meyrin and Prévessin | CERN Table Tennis Club | EN & FR
CERN 70th Anniversary Table Tennis Tournament

30 May 12:15 | At CERN | Meyrin site | CERN Running Club | EN & FR
52nd CERN Relay Race

30 May 14:30 | Knowledge Sharing | 513/1-024 and online | IT Innovation visits | EN

IEEE mission and activities in today's digital world

30 May 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN
Quantum Gravity and Predictions for our Universe

31 May 13:30 | Knowledge Sharing | 40/S2-C01 - Salle Curie | KT Seminars | EN
Global Health: Enhancing Access to Cancer Radiation Therapy

3–7 Jun | Physics | Main Auditorium | CERN, EPFL, ETHZ, LAPTh, Universities of Bern and Geneva | EN
Strings 2024 conference

4 Jun 12:00 (EN) 13:00 (FR) | At CERN | 40/S2-A01 - Salle Anderson | CERN psychologists | EN & FR
Work Well Feel Well Campaign: relieving daily stress / décharge quotidienne du stress

6 Jun 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN
CERN70 public event: The case of the (still) mysterious Universe (more CERN70 events)

7 Jun 10:00 | Knowledge Sharing | CERN Library | Library events | EN
Meet the authors - Shantha Liyanage, Markus Nordberg & Marilena Streit-Bianchi on "Big Science, Innovation and Societal Contributions"

10 Jun 14:00 | Knowledge Sharing | Council Chamber | TH department | EN
Alvaro at 80

10–12 Jun | Knowledge Sharing | Online | Sustainable HEP | EN
Sustainable HEP 2024 – 3rd Edition

11–13 Jun | Engineering | CERN | FDF | EN
1st FPGA (Field Programmable Gate Arrays) Developers' Forum meeting

13 Jun 17:30 | Knowledge Sharing | Council Chamber | Staff Association | FR
« La femme naît libre et demeure égale à l'homme en droits » , par Valentina Altopiedi

14 Jun 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
L&D Micro-talk 11 - Cultivating Emotional Intelligence (EI) in the Era of Artificial Intelligence (AI)

2 Jul–2 Aug | Knowledge Sharing | Main Auditorium | Summer Student Programme | EN
Summer student lecture programme 2024

2 Jul–31 Jul | Knowledge Sharing | IT Amphitheatre | CERN openlab | EN
CERN openlab summer student lecture programme 2024

4 Jul | 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN
CERN70 public event: Exploring farther – Machines for new knowledge

6 Jul 15:00 | At CERN | R3 terrace, Prévessin | Music Club | EN & FR
Hardronic Music Festival 2024

Registration now open

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
Introduction to Accelerator Physics

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events
Indico also shows ALL events happening today, this week and in a calendar view.
If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

SAVE THE DATE: 17 September CERN70 community event

The CERN community is warmly invited to a dedicated event to honour the Laboratory's 70th anniversary. This CERN community celebration will take place on Tuesday 17 September 2024 from 6 p.m. to 10 p.m. on the lawn next to Restaurant 1 on the CERN Meyrin site, and will include music, videos, food, drink and more.
The event celebrates the entire CERN community and is open to all CERN access pass holders, including family members and retirees. You will need to register to take part and registrations will open early summer. All attendees, including family members, will need to bring a CERN access pass to enter the event.

More details to come but, in the meantime, save the date of 17 September.

For now, you can still register to attend the 6 June event "The case of the (still) mysterious Universe" and find out more about the CERN70 celebration programme by checking cern70.cern.

Practical advice to release tension and relieve stress

Take part in the Work Well Feel Well campaign session with CERN psychologists on 4 June

On 16 April, HR announced the fourth theme of its Work Well Feel Well campaign, with the article "Release the pressure". In this context, the CERN psychologists invite you to a dedicated lunchtime session on 4 June, comprising both a theoretical and a practical component, with concrete techniques and tools to effectively relieve your stress in the moment.
The English session will take place from 12.00 p.m. to 12.45 p.m., followed by a session in French from 1.00 p.m. to 1.45 p.m. Please register to attend the event in person, which will take place in Building 40, Salle Anderson (40/S2-A01). The event will also be broadcast over Zoom and recorded.
https://indico.cern.ch/event/1392780/

You can also find more information and advice on the Medical Service's webpage dedicated to mental health support.

CERN psychologists

Glacier des Particules: the ice cream cart is back for summer 2024

Ice cream fans will be happy to hear that the Glacier des Particules ice cream cart has returned for summer 2024. It can be found on the Esplanade des Particules close to Building 33.

It has been there since 17 May and, weather permitting, will be in place every week from Tuesday to Saturday between 1 and 6 p.m. It will be open for four months, until mid-September

2024, and is sure to be a hit with both the CERN community and visitors to the site.
A wide selection of homemade ice creams is available at the price of 4 CHF for one scoop and 7 CHF for two.
Make sure you try one of their delicious ice creams or vegan sorbets!

HéLéman presents its carpooling solutions at CERN on 6 June

CERN has joined forces with two local partners, Mov'ici and HéLéman, to provide the CERN community with a free carpooling service.
The HéLéman network's cross-border carpooling service has been in place since 2019. In February 2024, a new carpooling route for the Pays de Gex was opened, serving various stops between Valserhône and Divonne-les-Bains.
If you would like to find out more about how passengers and drivers can connect in real time, where the different pick-up/drop-off points are located (no booking required) and how drivers are paid, come and meet the HéLéman team at CERN on 6 June.
Stands will be set up in Restaurants 1 and 2 from 10 a.m., and the HéLéman team will be handing out flyers at Gates B and E from 7.30 a.m.
Carpooling is a simple and effective solution for reducing the number of cars on the roads, limiting traffic congestion and cutting pollution. Why not give it a go?

SCE department

New charging station for private electric vehicles

As of 3 June 2024, the Site and Civil Engineering (SCE) department are making a charging station for private electric vehicles available to the CERN community.
The dedicated area is located just before the Meyrin border crossing (on the Swiss side) and can be accessed via the Chemin de Berne. A CERN access card is required to enter. The electricity will be supplied by the Geneva utility provider, SIG. The charging cost will be borne by the user in accordance with the SIG terms and conditions and the SIG-Move pricing system (see prices). These spaces may only be used for charging purposes. No parking is allowed.
The creation of these six charging points is part of the SCE department's overall action plan to provide the CERN community with sustainable transport options.

SCE department

Philip John Bryant (1942 – 2024)

Philip John Bryant. (Image: CERN)

Newly married and finishing his PhD from University College, London, Phil was recruited by CERN in November 1968 to work in the magnet group of the Intersecting Storage Rings (ISR) division, where his first task was to watch over the manufacture of the skew quadrupoles. The group, later renamed the Beam Optics and Magnets (BOM) group, was strongly involved in the commissioning and development of the collider, and Phil set up and tested a low-beta scheme, built from recuperated iron-core magnets, to validate this technology for the ISR, paving the way for the first superconducting low-beta insertion in a working accelerator. Later, he led the design and construction of the beamline from the PS that enabled ppbar collisions at the ISR, a development with which he became deeply involved. His name is also associated with coupling compensation, and more generally with the smooth operation of the collider until it closed in 1983.

A skilled communicator, Phil then moved on to assist Kjell Johnsen with setting up the CERN Accelerator School (CAS). He served as director of the school from 1985 to 1991, delivering many lectures himself and laying the foundations for it to become the valued institution that it is today. He then participated in a study of a B-meson factory for CERN before turning his attention to medical accelerators. Under his leadership this culminated in the Proton-Ion Medical Machine Study (PIMMS) of a synchrotron and its beamlines, which became the basis of the now operating medical centres for cancer treatment in Italy (CNAO) and Austria (MedAustron). In the early 2000s, Phil joined the LHC effort, serving as chair of the specification committee and taking responsibility for the contract office. Having to navigate deadlines, he shuttled between physicists, engineers, procurement officers and CERN’s legal team. In addition to his competence, Phil brought a contagious enthusiasm to the table, and would apply diplomatic skills in the conclusion of protocols with funding agencies and institutes.

On official retirement from CERN in 2007, Phil moved to Austria to be available for the medical facility under construction there. As this activity wound down, he increased his collaboration with the Vienna-based company CIVIDEC developing diamond radiation detectors, as well as continuing to improve the WinAgile program that he had developed for accelerator design, and lecturing – notably for CAS and the Joint Universities Accelerator School (JUAS).

Phil enjoyed scientific work – developing new ideas, working on his program and writing. He was the author of numerous papers, and his 2012 report on the advancement of colliders due to work done at the ISR is exemplary. A prodigious worker, he was nevertheless modest and always anxious to acknowledge the contribution of collaborators. Besides being a talented physicist and engineer, Phil was also good at drawing, his cartoons being especially appreciated. An inveterate "bricoleur," when not busy advancing accelerator technology, he was active with his hands at home.

Phil will be sorely missed. We grieve his loss, and offer our sincere condolences to his widow Sue, daughter Jackie and son Richard.

His friends and colleagues

Cédric Fournier (1981 – 2024)

It was with profound sadness that we learned of the sudden death of our colleague Cédric Fournier. Cédric started working at CERN at a young age, first as an employee of a contractor and later (from 2004 on) as a member of staff. He made significant contributions to the initial construction of the LHCb detector – one of the most demanding projects of his early career being the assembly of the 1500-tonne dipole magnet – and subsequently supported all installation, maintenance and upgrade activities for the LHCb experiment. He was a highly talented mechanical technician who enjoyed working on challenging projects, in particular large and complex mechanical structures, and his skills were universally appreciated within the entire collaboration. Over the years, he became the principal mechanical technician of the LHCb Technical Coordination team as well as the head of the LHCb workshop at the experimental site, and took on key responsibilities in terms of logistics and safety. The latter was particularly important to him. Not only was he responsible for protective equipment, but he also regularly took the initiative to ensure and improve safe work practices. His continuous efforts to improve the quality of life for everybody at the LHC Point 8 site were widely appreciated.

Cédric was a dynamic and caring person who was known for pursuing projects he believed in with great enthusiasm and determination. Whenever a colleague asked him for help of any kind, he was there; it is difficult to imagine our daily working life without him.

Cédric's sudden disappearance was a genuine emotional shock for all of us. While we did not have the chance to say goodbye to him, he will remain in our thoughts, and his achievements and contributions to LHCb will stay with us.

On behalf of the entire team, our deepest sympathies go to his family and friends. Thank you for everything, Cédric, and rest in peace.

His LHCb colleagues

We are deeply saddened to announce the death of Mr Cédric Fournier on 14 May 2024.

Cédric Fournier, who was born on 8 May 1981, worked in the EP department and had been at CERN since 2 October 2000.

The Director-General has sent a message of condolence to his family on behalf of the CERN personnel.

Social Affairs service
Human Resources department

Ombud's corner

Learning – or relearning – the art of dialogue

I took up my new functions as Ombud at the beginning of May, after 35 years in various administrative support and advisory roles at CERN. I would like to pay tribute to my predecessor in the role, Laure Esteveny, and to thank those who have recently called on my services for the trust they have placed in me.

I will take all the time needed to lend a sympathetic ear, to exchange ideas and to discuss possible ways of coping with whatever interpersonal problem, conflict or other issue they might be facing in the workplace. For example, I can help resolve interpersonal conflicts in situations where dialogue isn't working or has broken down altogether.

And this leads me to the following thought – in today's era of instantaneous communication, at any time and in any place, we're losing the habit of speaking face to face. So when disagreements arise, we have trouble, on the one hand, expressing how we feel or knowing where to start and, on the other hand, hearing what the other is saying.

Discussion, debate, negotiation and information exchange are the most common forms of communication in an organisation, and are extremely useful tools from many points of view. But dialogue is probably the most effective means of comprehending and tackling especially complex or controversial subjects.

Learning – or relearning – the art of dialogue when the weather is set fair, with the aim of maintaining high-quality professional relations, is a good way of preparing ourselves to cope with stormy weather that might be on the horizon. Conflicts can thus be the catalyst for healthy change or improvements.

However, since good intentions alone are not always enough, here are three principles to guide you in the art of dialogue:

1. Set aside time to discuss a subject of mutual interest, without an agenda, in a comfortable space and with the aim of creating a meaningful connection.
2. Practice active listening, which means listening until the end and asking questions with friendly curiosity without preconceived ideas.
3. Recognise the power of the spoken word – and of silence – during a discussion.

Dialogue is an art form that can sometimes require practice. Take inspiration from these three principles and see for yourself the positive changes they can bring about in your interpersonal relations and your ability to tackle tough situations.

Marie-Luce Falipou

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at https://mattermost.web.cern.ch/cern-ombud/.
The Ombud is available from Monday to Friday in office B500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at ombuds@cern.ch
More information can be found on the Ombud's website: https://ombuds.web.cern.ch/fr